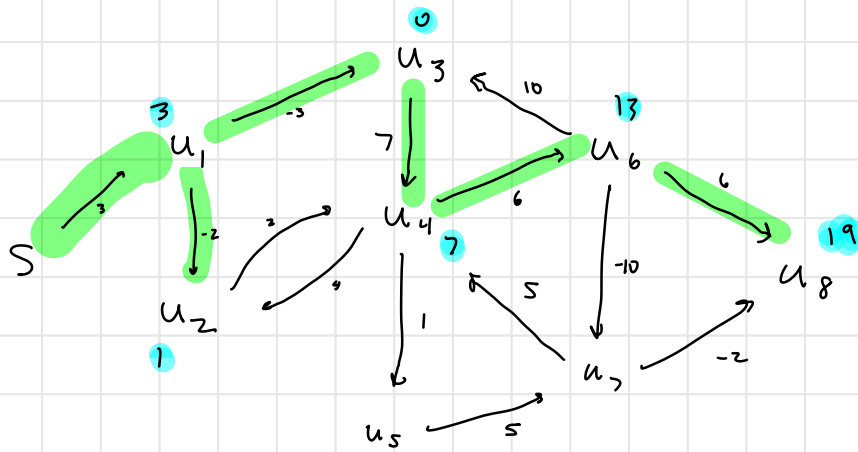


HW08

1)

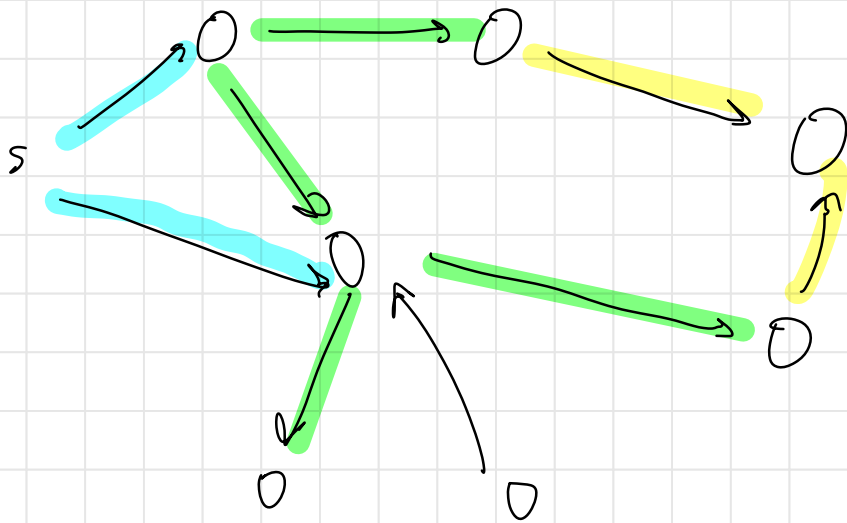


u_1/u_4 u_2/u_8 u_6/u_3 u_1/u_2 u_4/u_5 u_1, u_3
 u_3/u_4 u_6/u_5 u_4/u_7 u_2/u_7 u_1, u_7 u_4, u_2

2

worst case bellman - ford $\Theta(V \cdot E)$ no
early stops

$\Theta(\sqrt{V})$ edges to go through check
 $\Theta(E)$ times $\therefore \Theta(E \cdot \sqrt{V})$



● = pass 1

● = pass 2

● = pass 3

passes

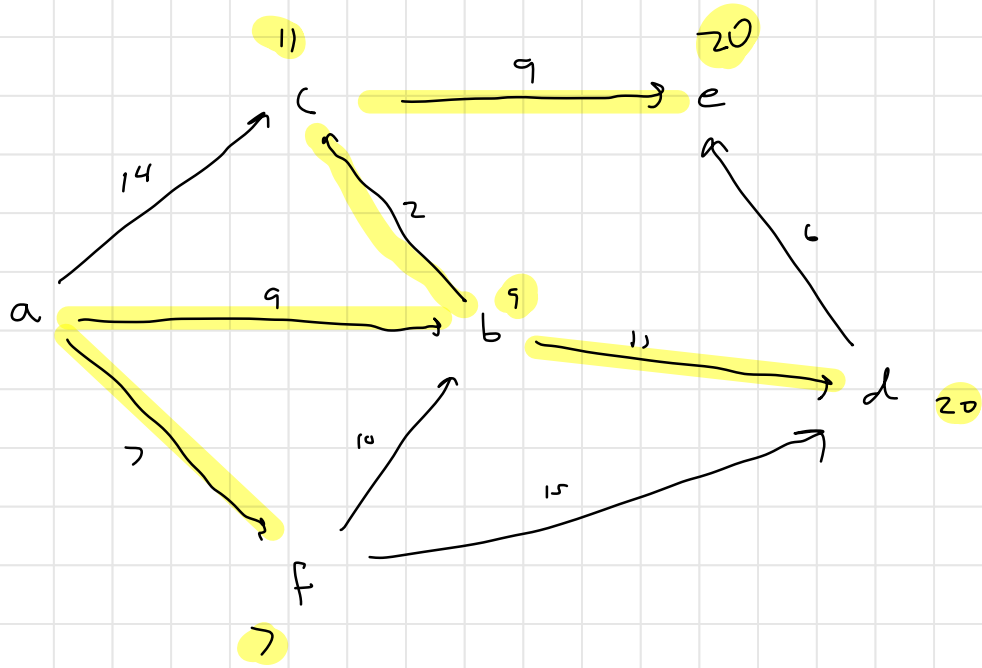
in the worst case

3 passes are needed

as the max step to
sub to all nodes is 3
in the worst case

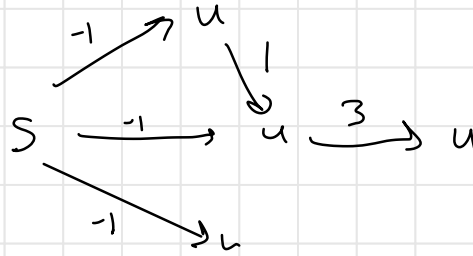
3 iterations of the outer loop ^{assuming} you only
process one
step in the worst
case each pass.

4



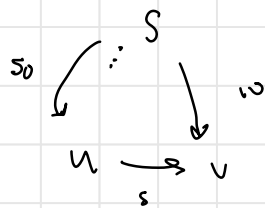
5

True



Since we are given that there is no negative loops only the edges from the source can be negative and the rest afterward will be positive. The algorithm can set all the paths from the source correctly, then all the updates can be made regularly afterwards.

6a) True,



$$S \rightarrow u \leq 50$$

$$S \rightarrow v \leq 10$$

$$S \rightarrow u \rightarrow v \leq 55 \quad \therefore S \rightarrow u$$

is shorter

Since the graph is directed if u goes to v only this does not make the path $\delta(S, u)$ any shorter.

6b) False, since the graph is undirected the shortest path from $S \rightarrow u$ or $S \rightarrow v$ can be the shorter of the two paths connect to u and v with a weight of 5

Since $\delta(S, v) \leq 10$ the $\delta(S, u)$ can be 15
(since $\delta(u, v) \leq 5$) instead to 50 that is
given